Phase 1 & 2: Identity, Auth & Sodia Interactions

Deep Dive 1: Auth & Onboarding Flow

EntryPoint: Splash Screen ➔ Auth Landing

- TAP **"Send OTP" (Phone Number Entered)**
  State: Transitions to OTP Verification screen. 6-box grid auto-focuses.
  - INPUT **Invalid 6-Digit Code**
    Error State: Boxes shake horizontally. Borders turn red. Toast: "Incorrect code."
  - AUTO-FILL **Valid 6-Digit Code via SMS**
    State: Instantly validates. Issues JWT.
    - SYSTEM **User is New**
      Route: Pushes to "Basic Info" (Avatar, Name, DOB).
      - INPUT **DOB under 13**
        Error State: "You must be 13 or older." Next button remains disabled.
      - TAP **"Next" (Valid Data)**
        Route: Pushes to Preferences (Goals). Seeding WRE algorithm.
        - TAP **"Complete Profile"**
          Result: Navigates to main Sodia Feed. Initialization complete.

Deep Dive 2: Sodia Feed & Stories

EntryPoint: Main Infinite Scroll Feed

- TAP **"Drop In" Pill on a Discovery Card**
  State: Universal Checkout bottom-sheet slides up over the feed.
  - TAP **"Confirm & Pay"**
    State: OS biometric prompt appears.
    - AUTH **Biometric Success**
      Result: Pass injected into Cards Hub. Sheet shows "You're in!" with "Share to Sodia" CTA.
- TAP **An Avatar in the Stories Header Tray**
  State: Opens full-screen Story Viewer. Playback auto-starts.
  - TAP **A Floating Location Tag (e.g. "Twins Fitness") inside the Story**
    State: Pauses story playback. Slides up the Checkout Bottom Sheet.
    - SWIPE DOWN **Dismiss Checkout Sheet**
      Result: Sheet closes. Story playback resumes automatically.

Phase 3 & 4: Discovery Map & VIP Wallet

Deep Dive 3: Discover Mapbox

EntryPoint: Discover Tab (Map Canvas)

- TAP **Top Search Bar**
  State: Search Overlay expands to 90% height. Keyboard focuses.
  - TYPE **"Pickleball"**
    Result: Debounced query groups results into Facilities, Challenges, and Users.
    - TAP **A Facility Result**
      Result: Overlay closes. Map instantly pans/zooms to that facility's pin and opens its bottom-sheet.
- TAP **Glowing Event Pin (Active Challenge)**
  State: Bottom sheet opens. Shows Facility details PLUS an "Accept Challenge" CTA.
  - TAP **"Accept Challenge"**
    Result: Silently injects a Bounty Card into the user's Track Tab. Button text changes to "Challenge Accepted".

Deep Dive 4: Meca Wallet (Cards & Ledger)

EntryPoint: Cards Tab (3D Stack)

- TAP **Center Card (Active Pass)**
  State: Card flips 180°. Screen brightness auto-maxes. Shows ticking 60s QR code hash.
  - TAP **QR Face Again**
    Result: Flips back to front face. Screen brightness returns to normal.
- TAP **"Send" Icon in Action Row**
  State: Opens "Select Friend" modal.
  - TAP **A Friend & Confirm**
    Result: Transfers `owner_id` in DB. Card instantly vanishes from user's stack and injects into friend's stack. SOA logs "Gift Sent".
- SCROLL UP **Bottom SOA Ledger**
  State: Ledger expands to full-screen, hiding cards. Search and Date filters become visible.
  - TAP **A Transaction Row**
    State: Opens Digital Receipt detail sheet.
    - TAP **"Get Help"**
      Result: Opens Support Chat, auto-attaching the `transaction_id` payload so agents have context.

Phase 5: Trinity Tracker (Lift & Move)

Deep Dive 5: Lift Engine (Active Grid)

EntryPoint: Lift Tracker Dashboard

- TAP **"Quick Start"**
  State: Opens Active Workout view. Live timer starts at 00:00.
  - TAP **"+ Add Exercise"**
    State: Opens Exercise DB sheet. Select "Bench Press" and close.
    - TAP **Status Checkmark on Row 1 (225 lbs x 8)**
      Result: Row turns green. Auto-saves draft. Floating Rest Timer pops up showing "01:30".
      - TAP **"+15s" on Rest Timer**
        Result: Timer adjusts to 01:45 without affecting the overall session duration timer.
    - TAP **"Finish" Workout**
      State: Confetti modal. Flags new PRs. Shows "Share to Sodia".
      - TAP **"Share to Sodia"**
        Result: Opens Camera/Gallery with uneditable Lift Stamp overlay. User can drag/resize stamp before posting.

Deep Dive 6: Move Engine (GPS & Court)

EntryPoint: Move Tracker Dashboard

- TAP **Universal Selector (e.g. "Trail Run")**
  State: Opens Setup Sheet. GPS toggle forced ON.
  - TAP **"Start" (GPS Enabled)**
    State: 3-2-1 countdown. Live Mapbox trail view starts.
    - LOCK **Device Screen (Mid-run)**
      Result: Background tracking ensures GPS array continues recording via OS permissions.
    - FORCE QUIT **App (Crash simulation)**
      Result: On reopen, Crash Recovery modal prompts "Resume active run?". Recovers last local cache.
- TAP **"Pickleball" (Court Mode)**
  State: Map vanishes. Replaced by pulsing HR ring and active timer.
  - TAP **"Finish"**
    State: Post-Match Prompt overlay.
    - INPUT **"Win" and Tag @opponent**
      Result: Completes workout. Sends notification to opponent to verify rivalry match result.

Phase 5 & 6: Nutrition, Profiles & Toggles

Deep Dive 7: Food Engine (Nutritionix NLP)

EntryPoint: Food Tracker Dashboard

- TAP **Search Bar**
  State: Keyboard focuses. Displays "Recent" and "Favorites" cache.
  - TYPE **"2 boiled eggs and 1 cup rice" (NLP Input)**
    Result: Queries Nutritionix. Parses into two distinct food items with correct macro splits.
    - TAP **"Log Items"**
      Result: Instantly updates Dashboard Top 4 Macro Widgets (Calories, Protein, Fat, Carbs).
  - TAP **Barcode Icon**
    State: Opens native camera.
    - SCAN **Unrecognized UPC**
      Error Route: Falls back to Manual Log modal ("Item not found. Please enter manually").
- TAP **"My Meals" Slider Pill (e.g. "Morning Oats")**
  Result: Instantly logs all 4 constituent ingredients to the daily ledger without requiring a detail confirmation.

Deep Dive 8: Profile Switching & Trophy Case

EntryPoint: User Profile Main Tab

- SWIPE **to Tab 2 (PRs & Entries / Trophy Case)**
  State: Renders grid of immutable Tracker Stamps (verified challenge attempts).
  - LONG PRESS **A Verified Entry**
    State: Context menu opens. Option: "Hide from Public". (Deletion is blocked to protect leaderboards).
- TAP **"Settings" Gear Icon**
  State: Opens Global Settings.
  - TAP **"Switch to Business Mode" (If Staff)**
    Result: Instantly tears down B2C navigation stack and mounts the GymDash B2B Fast-Lobby interface.
  - TAP **"Integrations" -> "Connect Garmin"**
    Result: OAuth flow. Enables background polling for HR Zone (Z1-Z5) charting in the Move Tracker.

B2B Integration: GymDash & CoachDash

Deep Dive 9: GymDash Web OS & Tablet

EntryPoint: Staff Clock-In & Tablet Gate

- TAP **"Clock In" on Personal Sodia App (Staff Mode)**
  State: Opens Selfie Cam. Checks GPS.
  - CAPTURE **Photo within Geofence**
    Result: Staff clocked in for HR/Payroll. DOES NOT grant POS access yet.
- TAP **Till on GymDash Tablet**
  State: Demands PIN.
  - INPUT **PIN**
    State: Float Gate active. Demands physical count input.
    - INPUT **Opening Cash Amount**
      Result: Unlocks POS. Cash liability is now bound to that Staff ID until shift handover.
- SCAN **User Sodia QR at Gate (Tablet Feed)**
  State: Validates JWT hash against local cache.
  - SYSTEM **Valid / Expired Checks**
    Good Result: Flashes GREEN. Increments "Inside Now" dynamic capacity counter.
    Bad Result: Flashes RED. Shows reason code (e.g. "Pass Expired"). Prompts Staff to hit "Manager Override" or sell renewal.

Deep Dive 10: CoachDash CRM & Organizer Pro

EntryPoint: Coach & Organizer Portals

- VIEW **Coach Triage Dashboard (Morning)**
  State: System calculates variance between `planned_track` and client's `actual_track` from Sodia.
  - SYSTEM **Client Missed Macros for 3 Days**
    Result: Auto-triggers an automated check-in push notification to the client's Sodia Inbox to secure retention.
- TAP **"Create Event" in Organizer Pro (Funnel Builder)**
  State: 3-Step Wizard starts.
  - COMPLETE **Venue Handshake (Step 2)**
    Result: Sub-division spatial engine locks "Pickleball Court A", while leaving Courts B & C available in the Sodia B2C Discover map.
    - TAP **"Close Registration & Generate Bracket"**
      Result: JSON bracket engine ingests roster, historically seeds teams based on Sodia player ratings, and pushes interactive tree to all users.

Automated Routing & B2B Yield Management

Deep Dive 11: The Waitlist Sniper Engine

EntryPoint: Sodia Checkout (Fully Booked Class/Court)

- TAP **"Join Queue" (Slot is Full)**
  State: Opens Checkout Sheet in Pre-Auth mode.
  - AUTH **Confirm & Pre-Authorize Payment (Biometric)**
    Result: Payment gateway places a hold. DB injects user into `waitlist_queue`. Screen shows "You are #1 in queue."
    - SYSTEM **Original Booker Cancels their Slot**
      State: System cron job intercepts cancellation. Reads `waitlist_queue`.
      - SYSTEM **Auto-Capture & Route**
        Result: Xendit captures the pre-auth hold. System generates QR pass, injects to Cards Hub, and sends Push Notification: "You're in! Your waitlist cleared for [Facility]."

Deep Dive 12: Squatter Detection & B2B Upsells

EntryPoint: GymDash QuickDash Tablet (Gate)

- SYSTEM **14 Users Scan QRs Linked to a Single 10-Person Court Booking**
  State: Capacity threshold breached. Triggers WebSocket alert to Desktop Web OS.
  - VIEW **"Needs Attention" Queue on Desktop OS**
    State: Flashes RED: "Capacity Breach: Court 1 (14/10)".
    - TAP **The Alert Row**
      State: Opens Detailed Scan Log identifying the 4 unauthorized scans.
      - TAP **"Force Upsell to Organizer"**
        Result: Silently pushes an invoice payload via Sodia DM to the Booking Owner (Organizer) for 4 extra "Drop-In" fees.

Media Compositing & B2B CRM Integrations

Deep Dive 13: Tracker Stamp Compositing (Sodia Harness)

EntryPoint: Tracker Complete Modal (Lift/Move/Food)

- TAP **"Share to Sodia"**
  State: Mounts Camera Viewfinder/Gallery modal. Uneditable SVG Stamp (Data Payload) floats on top.
  - PINCH / DRAG **The Stamp**
    Result: Scales and translates the stamp coordinates relative to the screen bounds.
    - TAP **The Stamp Body**
      Result: Inverts CSS filter from Light Mode (white text) to Dark Mode (black text) for contrast.
    - TAP **"Post to Story"**
      Result: Cloud function composites the layer, publishes to Stories (24h expiry), AND auto-pins the raw stamp to Profile Tab 2 (Trophy Case) immutably.

Deep Dive 14: Predictive Attrition (Churn Win-Back)

EntryPoint: GymDash Desktop OS (Members Tab)

- SYSTEM **Nightly Cron Job Evaluation**
  Result: Identifies User A (Pass expires in 7 days AND 0 check-ins in last 14 days). Moves User A to "At-Risk" queue.
  - TAP **User A in "At-Risk" Queue**
    State: Opens Member CRM Profile.
    - TAP **"Send Win-Back Offer"**
      State: Modal opens: Select 10%, 20%, or 30% discount SKU.
      - TAP **"Confirm & Push"**
        Result: Injects an interactive "Discount CTA Pill" directly into User A's Sodia Direct Messages. If User A taps it, it routes straight to biometric checkout.

Account Safeguards & Competitive Validation

Deep Dive 15: Account Deletion Constraints

EntryPoint: Sodia App -> Settings -> Account

- TAP **"Delete Account"**
  State: Demands High-Friction FaceID/Biometrics immediately.
  - SYSTEM **Queries `passes` table for active entitlements**
    Result: Locates 1 active Monthly Subscription.
    - SYSTEM **Hard Block Triggered**
      Error State: Modal pops: "You have active gym passes. Please cancel subscriptions or wait for expiry before deleting to prevent financial lock-outs."
      - TAP **"Deactivate Instead" (Secondary Option)**
        Result: Soft-deletes profile. Hides user from WRE, Maps, and Search, but keeps ledger math intact for the B2B facility.

Deep Dive 16: Move Tracker (Court Mode Post-Match)

EntryPoint: Move Tracker (Pickleball/Tennis Mode)

- TAP **"Finish Match"**
  State: Opens Post-Match Prompt Overlay.
  - INPUT **"Win" & Score (e.g. 11-8)**
    State: Expands tag input.
    - INPUT **Tag @OpponentUsername**
      Result: Generates Move Stamp. Pushes notification to Opponent.
      - ACTION **Opponent Views Notification**
        State: Sodia DM shows "Match Verification Request: Loss (8-11)".
        - TAP **"Confirm Result" (by Opponent)**
          Result: Writes verified match to DB. Updates the Sodia "Rivalry Ledger" between the two users, posting to feed automatically.

Offline Resilience & B2B2C Routing

Deep Dive 17: Offline Ledger Sync Protocol (GymDash + Sodia)

EntryPoint: Sodia App & GymDash (No Network Connection)

- TAP **Center VIP Card (User offline)**
  State: App detects no network. Cannot fetch dynamic QR hash.
  - SYSTEM **Fallback Triggered**
    UI Update: Generates a 6-digit TOTP code on screen (e.g. "482-911") with 60s countdown.
    - INPUT **Staff types TOTP into Offline GymDash Tablet**
      State: Validates code against local SQLite `active_passes` cache downloaded at shift start.
      - SYSTEM **Validation Success**
        Result: Flashes GREEN. Logs event to `local_sqlite_queue` with unique idempotency key.
- SYSTEM **Network Reconnection (Background)**
  Route: WebSocket reconnects. GymDash pushes `local_sqlite_queue` payload to Supabase `ledger_transactions`.

Deep Dive 18: B2B2C Financial Split (Coach Floor Fees)

EntryPoint: CoachDash ➔ Session Booking

- TAP **"Confirm Booking" (At Registry A Gym with Floor Fee)**
  State: System deducts 1 session from Client's active package.
  - SYSTEM **Stripe Connect Trigger**
    Result: Deducts ₱150 (Floor Fee) from Coach's digital wallet and routes instantly to Gym's merchant ledger.
    - SCAN **Client Arrives & Scans Sodia QR at GymDash Gate**
      System Logic: Reads `entitlements` table. Detects no base gym membership, BUT detects valid `coach_session_id`.
      - SYSTEM **Dual-Authorization Success**
        Result: Gate flashes BLUE (Coach Client). Grants turnstile access. Sends WebSocket ping to CoachDash: Client status flips to [IN BUILDING].

Data Migration & The Algorithm Engine

Deep Dive 19: Legacy CSV Onboarding (GymDash to Sodia)

EntryPoint: GymDash Web OS ➔ Data Import Wizard

- UPLOAD **Legacy CSV (Name, Phone, Pass Expiry)**
  State: Parses CSV into `offline_members` table (no Sodia profile yet).
  - SEARCH **Walk-in Member via Phone Number on Tablet**
    Result: Matches offline record. Shows "Valid until Dec (Offline)".
    - TAP **"Check-In & Digitize"**
      Result: Prints physical receipt with Claim QR Code.
      - SCAN **User scans Claim QR via Sodia App**
        Route: Completes OTP verification. Maps `phone_hash` to `claim_code`. Converts offline member to `public.profile` and injects VIP Pass into 3D Wallet.

Deep Dive 20: Gamification & WRE Multipliers

EntryPoint: Track Engine ➔ Sodia Feed Assembly

- TAP **"Save Workout" (e.g. Powerlifting)**
  State: Writes to `workout_logs`. Checks consecutive days for Streak updates.
  - SYSTEM **WRE Cron Job (Background)**
    Logic: Detects 3 powerlifting logs this week. Updates `user_weights.powerlifting` multiplier from 1.0 to 1.5.
    - VIEW **User opens Sodia Feed (Next Launch)**
      Route: WRE injects Discovery Card at Slot #4. Queries facilities/challenges matching `powerlifting` tag based on new multiplier weight.

Registry B: Clubdash & Organizer Pro

Deep Dive 21: Organizer Pro ➔ Mobile Funnel Builder

EntryPoint: Organizer Pro App ➔ [+ New Event]

- SELECT **Sport [Basketball], Format [3v3], Bracket [Elimination]**
  State: Transitions to B2B Venue Handshake (Step 2).
  - SELECT **Metro Sports Arena ➔ Saturday 8AM - 5PM**
    System Logic: Queries Clubdash `master_calendar`. Locks block as [Pending].
    - TAP **"Lock Venue & Agree to Split" ➔ Set Price ➔ "Publish"**
      Result: B2B Smart Contract executes. Generates landing page and pushes Tournament Card to Sodia Discovery feeds. Scarcity engine engages (auto-surges price of final 2 slots if toggled).

Deep Dive 22: Clubdash ➔ Spatial Sub-Division & Squatters

EntryPoint: Facility Web OS (Command Center)

- SYSTEM **Parent Asset [Basketball Court 1] Booked by Organizer**
  System Logic: Automatically blocks out Child Assets [Pickleball Courts A, B, C] for that exact window. Injects 15-min changeover buffer at 5:00 PM before reopening for public Sodia booking.
- SCAN **Scan #51 at Quickdash Gate for a 50-Person Event**
  State: System cross-references `quickdash_scans` vs Organizer's `booking_capacity`. Capacity breached.
  - SYSTEM **Alert Triggered**
    UI Update: Web OS and Tablet flash RED: "Alert: 51 scans for 50-person event."
    - TAP **"Enforce Tier 0 Base Waiver" (Staff)**
      Result: System forces the 51st user to sign a digital waiver and pay a base micro-transaction directly to the facility on their Sodia app before granting entry.

Game-Day Execution & Settlement

Deep Dive 23: Organizer Pro ➔ Game-Day Command Center

EntryPoint: Organizer Pro App (Live Action Mode)

- **SYSTEM** **Registration Closes (Bracket Generation)**
  Result: JSON engine ingests roster, historically seeds teams via Sodia ratings, and generates visual tree.
  - **INPUT** **Final Score (e.g. 21-18) ➔ Tap "Finalize"**
    System Route: Updates JSON bracket globally. Advances winner, eliminates loser.
    - **SYSTEM** **Automated Push Notifications**
      Result: Pushes alert to winner: "You won! Next match in 15 mins." Pushes alert to next opponent: "Your opponent is ready."
    - **TAP** **"End Tournament"**
      System Route (Stripe Connect): Calculates gross revenue. Deducts platform fee. Routes Venue Rental Fee to Facility bank account. Routes net profit to Organizer digital wallet. (Zero manual invoicing).

Edge Cases, Concurrency & System Fallbacks

Deep Dive 24: Concurrency "Race Condition" & Waitlist Sniper

EntryPoint: Universal Checkout & Waitlist Engine

- TAP **"Confirm & Pay" (User A - 1 slot left)**
  State: System places a strict 3-minute Pessimistic Lock on the inventory row in the database.
  - CONCURRENT ACTION **User B taps "Confirm & Pay" (Same millisecond)**
    System State (User B): Transaction blocked by DB lock. UI shows: "Slot just taken! [Join Waitlist]".
  - SYSTEM **User A's Biometric or Network Fails (Timeout)**
    Result: 3-minute lock expires. Slot returns to the public pool automatically.
- SYSTEM TRIGGER **User A cancels their court booking 2 hours before the event**
  Route: System queries the `waitlist_array`. DOES NOT broadcast to all users to prevent a stampede.
  - SYSTEM **Route to Waitlist #1**
    UI Update: Push Notification to Waitlist #1 User: "A spot opened up! You have 15 minutes to claim it."
    - ACTION **Waitlist #1 Ignores or Misses Push**
      Route: 15-minute cron job expires the exclusive offer. Generates new push to Waitlist #2 User.

Deep Dive 25: GDPR vs. Immutable Ledger (Data Deletion)

EntryPoint: Sodia Settings ➔ Data Compliance

- TAP **"Delete Account" ➔ Enter FaceID**
  State: Account enters 30-day "Soft Delete" grace period. UI logs out user.
  - SYSTEM TRIGGER **Day 31 Hard Delete Cron Job Executes**
    Route (B2C): Wipes `public.profiles` (Name, Phone, Bio, Avatar, Auth credentials).
    - SYSTEM ROUTE **GymDash B2B Ledger Integrity**
      Logic: Identifies all `ledger_transactions` tied to the user's UUID.
      - MUTATION **Anonymization Protocol**
        Result: Reassigns buyer name to "Anonymous / Deleted User". PRESERVES `transaction_id`, date, and monetary amount. GymDash Gross Sales reports remain 100% mathematically accurate.

Financial Reversals & Media Pipelines

Deep Dive 26: Cascading Refund & Void Routing

EntryPoint: Organizer Pro App ➔ Event Management

- TAP **"Cancel Tournament" ➔ Select Reason [Weather/Rain]**
  State: Prompts confirmation: "This will refund all 16 teams. Proceed?"
  - TAP **"Confirm"**
    Route: Stripe API triggers a cascading split-clawback.
    - SYSTEM **Step A: Organizer Clawback**
      Result: Claws back Net Profit from Organizer's digital wallet. (Fallback: Stripe issues negative balance if funds were already withdrawn).
    - SYSTEM **Step B: Facility Clawback**
      Result: Claws back the Venue Rental Fee from the Clubdash Facility Owner.
    - SYSTEM **Step C: Consumer Refunds**
      Result: Consolidates funds and issues full refunds to 16 Sodia users. Sodia Cards Tab Ledger row updates from "Paid" to "Refund Processing".

Deep Dive 27: Media Compositing Pipeline (Stamp Editor)

EntryPoint: Sodia Harness ➔ Post Creator

- DRAG / PINCH **"Verified PR" Graphic Stamp over 4K Video**
  State: UI temporarily stores X/Y coordinates, scale multiplier, and rotation degrees in local state.
  - TAP **"Post"**
    Route: Foreground/Background FFMPEG render pipeline initiates.
    - SYSTEM **Frame Extraction & Flattening**
      Action: Extracts frames, flattens the SVG/PNG watermark onto the video frames using the saved coordinate state.
    - SYSTEM **Compression & CDN Upload**
      Result: Compresses output to 720p H.264 (preventing OOM crashes). Uploads payload to Supabase Storage. Returns CDN URL, writes to `posts` table, and deletes local temp file.

Extreme Hardware & OS Fallbacks

Deep Dive 28: Hardware Disasters & Offline Saves

EntryPoint: GymDash QuickDash & Sodia App

- CONTEXT **Tablet Camera Failure (GymDash)**
  State: Front-desk iPad camera shatters. Scanner is dead.
  - TAP **"Manual Lookup / Keyboard" Icon**
    State: Opens text input. Staff types User's Phone Number or Name.
    - TAP **"Manual Check-In" on User Profile**
      Result: Bumps "Inside Now" capacity. Records entry method as `manual_staff_override` in DB for audit logs to prevent insider abuse.
- CONTEXT **Phone Dies Mid-Run (Sodia Move Tracker)**
  State: User is 8km into 10km run. Battery dies. Background GPS tracking halts. SQLite `active_run` row remains unclosed (`status: in_progress`).
  - ACTION **User charges phone & opens Sodia App**
    Route: App boot checks local SQLite for unclosed runs.
    - SYSTEM **Prompt Overlay**
      UI: "Your device powered off during a run. Recover partial data?"
      - TAP **"Recover & Save"**
        Result: Closes the run at the exact timestamp and GPS coordinate of the battery death. Generates the Move Stamp for the 8km achieved.

Identity Context & POS Tendering

Deep Dive 29: Identity Switching & Context Payload ("Post As")

EntryPoint: Sodia Main Feed ➔ Create Story Modal

- TAP **"+ Your Story" on the header tray**
  State: Full-screen viewfinder opens. Top bar displays the `[Post As: Individual]` toggle.
  - TAP **"Post As: Individual"**
    State: Bottom sheet opens showing available roles mapped from `public.user_roles`.
    - TAP **"Staff: Twins Fitness"**
      Route: Context payload updates. `author_id` and `author_type` reassigned to facility.
      - CAPTURE & POST
        Result: Inserts story with facility as publisher. Bypasses user's personal followers, pushing to facility's followers.

Deep Dive 30: GymDash POS — Mixed Cart Tendering

EntryPoint: GymDash Tablet ➔ POS Till

- SELECT **Day Pass (Access) + Water Bottle (Store)**
  State: Cart shows 2 items. Total updated.
  - TAP **"Who is buying?" ➔ Select "Quick Cash / Walk-in"**
    System Logic: Detects `Access` SKU (platform fee applies) mixed with `Store` SKU (100% gym cash).
    - SYSTEM **Gateway Disable**
      UI Update: `[Pay via App]` dynamically disables to protect ledger routing.
      - SELECT **Cash or External Terminal**
        Result: Posts to append-only ledger, issues digital entitlement, and prints physical receipt with Claim QR.

Data Versioning & Sync Conflicts

Deep Dive 31: GymDash SKU Engine — Price Updates

EntryPoint: Facility Web OS ➔ Data Values & Entries

EDIT **Base Price of Existing Sub (e.g. ₱1,500 to ₱1,700)**

State: User taps "Publish".

SYSTEM **Immutability Guardrail**

Result: System DOES NOT overwrite `price` column. Duplicates row to `[Monthly Sub V2 - ₱1,700]`.

SYSTEM **Update V1 Status**

Result: Changes `[Monthly Sub V1]` status to `hidden_from_new_sales` but leaves active for current subscribers. Facility Sodia profile shows ₱1,700 for new checkouts.

Deep Dive 32: Track Engine — Wearable Conflict Detection

EntryPoint: Background Wearable Sync

SYSTEM **Detects new payload from Garmin Connect**

System Logic: Queries `actual_track` for overlapping start/end timestamps. Detects manual entry overlap.

SYSTEM **Flag Conflict**

UI Update: Pushes local notification: "New run detected. Tap to resolve sync."

TAP **Notification**

State: Modal opens: "Overlapping activity found. Which to keep?" (App GPS vs Garmin).

SELECT **"Keep Garmin Watch Data"**

Result: Overwrites manual entry with high-fidelity Garmin time-series data, enabling HR Zones (Z1-Z5).

B2B Handshakes & Payload Injections

Deep Dive 33: B2B Staff Handshake & POS Pin Issuance

EntryPoint: Facility Web OS ➔ Team Management

- TAP **"+ Invite Staff" ➔ Enter @username ➔ Send**
  Result: Pushes official B2B Handshake Card to staff member's Sodia Inbox.
  - TAP **"Review & Accept" (Staff Side Sodia)**
    State: Terms modal opens. Staff taps "Accept Role".
    - SYSTEM **Role Update**
      Result: Updates `public.user_roles` array. UI prompts to create 4-Digit POS PIN.
      - INPUT **4-Digit PIN**
        Result: Saves PIN to authorized staff table. Enables staff to pass Float Gate on GymDash tablet.

Deep Dive 34: CoachDash — Macrocycle Injection

EntryPoint: CoachDash ➔ Cycle Builder

- DRAG **Workout Block onto Client Calendar ➔ "Publish"**
  Route: Writes schema to client's `planned_track` table.
  - VIEW **Client opens Sodia ➔ Track ➔ Lift**
    State: Dashboard renders `[Active Program]` card mapped directly from Coach's JSON payload.
    - TAP **"Start Workout"**
      Result: Loads active workout engine grid pre-filled with coach's exact sets/reps/RPE. Failed targets auto-log to `actual_track` and flag Coach CRM.